

Research Highlights

Manuscript Title: A High-Resolution Finite-Difference Framework for Steady Continuation and Finite-Window Unsteady Dynamics up to $Re = 100,000$ in 2D Lid-Driven Cavity Flows

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- Exact $\mathcal{O}(N^2 \log N)$ DST-I Poisson solver eliminates iterative residual drift.
- Pure central differencing retains zero numerical viscosity for $Re \leq 50,000$.
- Interior hybrid scheme stabilizes corner singularity at $Re = 100,000$.
- Core vorticity profiles show qualitative alignment with Batchelor theorem.
- Unsteady marching on 513×513 grids resolves vortex shedding dynamics.